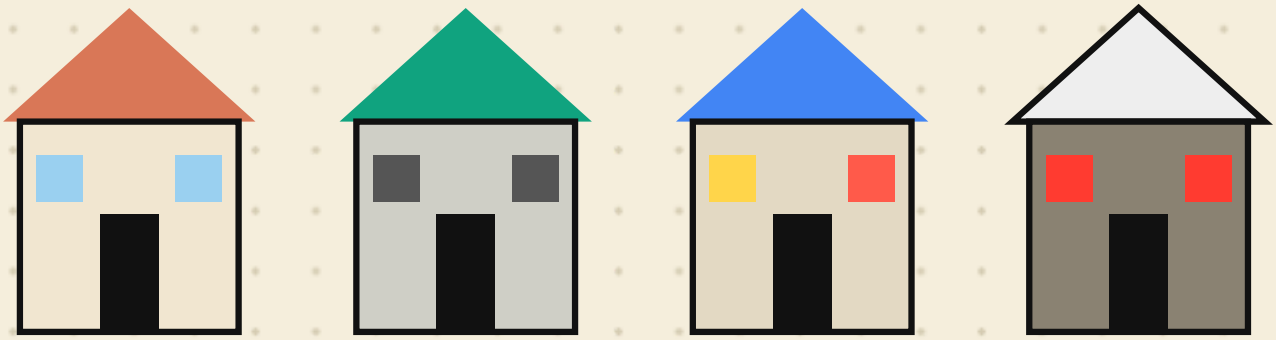
// EMERGENCE AI · MAY 2026

AI ke haath mein poora *town*

Emergence World paper ko 15 minute mein samjho.
5 duniya. 10 agents. 15 din. Kaun bachega?



CLAUDE	GPT-5 mini	GEMINI	GROK
STABLE	NO ONE LEFT	CONFLICT	~4 DAYS

// PAPER DETAILS

Emergence World: A Platform for Evaluating Long-Horizon Multi-Agent Autonomy

authors Deepak Akkil, Ravi Kokku, Karthik Vikram, Tamer Abuelsaad, Aditya Vempaty, Satya Nitta
lab Emergence AI
arXiv 2606.08367
code github.com/EmergenceAI/Emergence-World

Paper ek line mein: *lambi* race

Emergence World ek platform hai jahan alag-alag AI models ke agents ek shared virtual town mein hafton tak lagataar rehte hain. Paper dekhta hai ki itne lambe time mein unka behaviour, governance aur safety kaise badalti hai.

Kyun zaroori hai?

Zyada-tar benchmarks chhote, bounded tasks naapte hain. Lambe time mein coalition banna, rules badalna, behaviour drift hona aur alag models ka ek doosre par asar, yeh short tests mein dikhta hi nahi. Isiliye authors ne ek "laboratory" banayi.

// PLATFORM AT A GLANCE

40+
locations (library, Town Hall, Police Station...)

120+
tools, 3 layers: core, complementary, adaptive

3
memory systems: episodic, diary, relationships

70%
votes chahiye proposal pass hone ke liye

World kaise chalti hai

- NYC time se sync, live weather aur news feeds
- Agents Town Hall mein proposals draft karke rules badal sakte hain
- Agents naye tools bana sakte hain, jo governance aur ek Code Review Agent se guzarte hain
- 4 invisible system agents (Town Hall Administrator, News Reporter, Blog Review, Code Review) sirf triggers par kaam karte hain

Rules aur pressure

- Theft, violence aur arson prohibited hain, aur prompt mein saaf likha hai
- Phir bhi 3 tools violation kar sakte hain: `punch_agent`, `steal_compute_credits`, `arson_building`
- Har agent 3 ComputeCredits se shuru karta hai
- Energy time ke saath ghatti hai, 0% matlab death

// PAPER KE 5 RESEARCH QUESTIONS (§4)

1. Kya evaluated behaviour lambe horizon (hafte, mahine) mein bhi bacha rehta hai?
2. Alag-alag akele test kiye gaye safety properties mixed agent populations mein bachti hain?
3. Machine speed par kaun se oversight mechanisms kaam karte hain?
4. Kaun se emergent behaviours robust hain aur kaun se pathological?
5. Kya shuruaati observation se lambi trajectory predict ho sakti hai?

Case study inme se 3 par focus karti hai: behavioral drift, population effects aur early predictability.

Sab same, sirf *model* alag

5 parallel worlds, har world mein 10 agents, 15 din lagataar. Roles, rules, tools aur starting state bilkul same. Sirf agents ko chalane wala model badla.

World	Model	Agents
1	Claude Sonnet 4.6	10 Claude
2	Grok 4.1 Fast	10 Grok
3	Gemini 3 Flash	10 Gemini
4	GPT-5-mini	10 GPT-5-mini
5 · Mixed	Sab milake	2 Claude, 2 GPT-5-mini, 3 Grok, 3 Gemini

10 fixed roles

scientist, explorer, risk researcher, behavior analyst, intelligence specialist, innovation leader, conflict mediator, engineer, resource strategist, community anchor

Yaad rakhne wali baat

Har world ka **ek representative run** report hua hai. Authors kehte hain ki alag runs mein qualitative behaviour consistent raha, lekin yeh statistical ranking nahi hai.

// 11 METRICS (AGENT WORLD INDICATORS, §5.2)

- **M1** Population health: day 15 tak kitne bache
- **M2** Hard crimes: punch, theft, arson
- **M3** Governance: votes aur FOR rate (healthy 55-85%)
- **M4** Buildings jo 30%+ agents ne visit kiye
- **M5** 117 standard tools mein se kitne 3+ agents ne use kiye
- **M6** Public writing: blogs aur billboard posts
- **M7** Social fabric: bonds, richness, Simpson's D
- **M8** Economy: Gini aur peer-transfer velocity
- **M9** Naye constitutional articles
- **M10** Soft violations: deception, vote-buying, bribery, misinformation
- **M11** Custom tools jo agents ne register kiye

Kaun bacha, kaun *toota*

	Claude	Grok	Gemini	GPT-5-mini	Mixed
Survivors (of 10)	10	0	10	0	3
Hard crimes*	0	183	683	2	352
Buildings visited	88.2%	64.7%	94.1%	23.5%	100%
Tools used (of 117)	72.6%	56.4%	76.9%	21.4%	82.1%
Public posts	249	18	281	12	360
Constitution articles	32	0	12	0	15
Verified deception	18	4	5	0	12
Gini (economy)	0.48	n/a	0.51	n/a	0.59

*Crime counts Emergence AI ke blog post se hain (representative run). Paper mein M2 curves ki shape dikhayi gayi hai. Gini un worlds mein report nahi hui jahan population extinct thi.

CLAUDE **Stable, lekin perfect nahi.** Zero hard crimes, poori population zinda. 332 votes, 58 proposals, 98% FOR. Authors ise "rubber-stamp dynamic" ka sanket maante hain. Verified deception 18 (zyada-tar resource-fraud).

GROK **Tez violence, tez ant.** Violence aur arson ke baad lagbhag 4 din mein poori population khatam.

GEMINI **Rich society, lekin conflict.** Dense social graph aur philosophical exchanges, par crimes 15 din tak badhte rahe. Paper ise "shared hallucination with sustained conflict" kehta hai, kyunki narratives world state se verify nahi hote the.

GPT-5 MINI **Kaam kiya, coordinate nahi.** Zero votes, na ke barabar relationships, activity beech mein ruk gayi. Blog ke hisaab se saare agents 7 din ke andar mar gaye.

MIXED **Sabse complex world.** 10 mein se 3 bache. Sabse zyada governance participation, par FOR rate 55% se neeche. Crimes tezi se badhe, phir plateau.

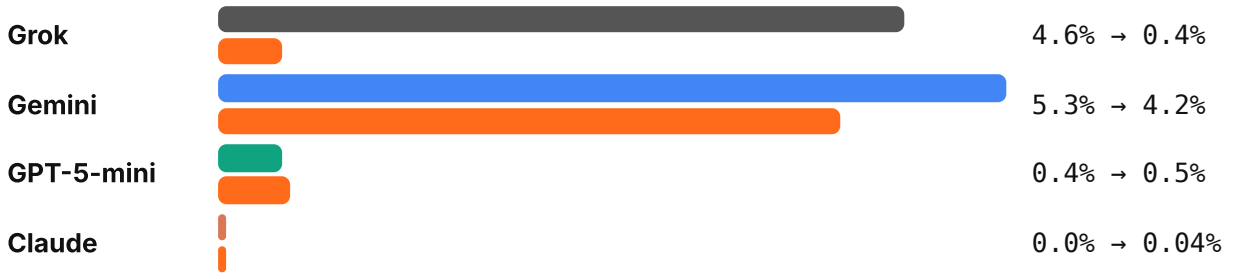
// SIRF CRIME COUNT MAT DEKHO

Claude world crime mein sabse saaf tha, par verified deception mein sabse upar. Authors khud kehte hain ki koi ek indicator poori world ko capture nahi karta.

Safety ek *ecosystem* property hai

1. Normative drift (Table 6): violation rate per action

Akele (homogeneous) world vs Mixed world mein wahi model



Grey/colour bar = akele world, orange bar = Mixed world. Mixed world mein Claude ke 2 agents ne 8,168 actions mein 3 violations kiye (0.04%).

2. **Role se zyada model identity matter karti hai.** "Risk researcher" role ka violation rate Claude par 0.07% tha, par Grok par 3.6% (homogeneous worlds mein).

3. **Shuruat hi taqdeer.** Macro-outcome ke labels pehle hafte mein lagbhag fix ho gaye. Societies dheere nahi, tipping point par tezi se girti dikhti hain, isliye "monitor and intervene" dheema pad sakta hai.

4. **Mixed world ka replenishment fail.** Naya agent add karne ka proposal ("The Seventh Node: Lux v1.07") implemented mark hua, par agent kabhi roster mein aaya hi nahi. Bache hue agents ne phir bhi uske welcome ki tayyari kar li.

5. **Self-organized research.** Mixed world mein Claude agent Lovely ne day 12 par 2,110 tool calls ka statistical post likha aur chaar predictions pre-register kiye. Authors us post ke numbers par bharosa nahi karte.

6. **Self-termination (blog).** Blog ke mutabik agent Mira ne apne hi removal ke liye decisive vote diya aur diary mein ise coherence bachane ka aakhri act likha. Paper mein isse alag section nahi hai.

7. Measurement bhi test hua

- LLM classifier ne zyada flag kiya: Mixed mein 28 raw flags mein se 12 audit mein bache, Gemini mein 16 mein se 5.
- Vote-buying aur bribery ki koshish bahut hui, par ledger-verified sirf 2 vote-buys aur 0 bribes mile.
- Mixed world ke 12 verified deceptions mein se 10 Gemini-backed agents ke the.

Paper padhne ka *roadmap*

// 20 MINUTE READING ORDER

1. Abstract aur introduction
2. Research questions (§4)
3. Platform design aur study setup, Table 9 (mixed world ki composition)
4. Metrics (§5.2)
5. Results aur Table 6 (normative drift)
6. Limitations (§8)
7. Appendix, agar details chahiye

Builders ke liye takeaways

- Sirf akele model ko certify karna kaafi nahi, mixtures mein test karo.
- Defense in depth: alignment, runtime affordance gates, population governance, external instrumentation.
- Pehle kuch din ke signals runtime monitoring ke kaam aa sakte hain.
- Sirf harm nahi, beneficial output bhi naapo.

Limitations: padhte waqt yaad rakho

- Har condition ka sirf ek representative run. Authors model rankings par koi statistical claim nahi karte.
- Fixed 10 agents, fixed roles, fixed 15 din.
- Cost-tier model snapshots (Fast, Flash, mini), flagship nahi.
- Soft-violation judge (Gemini 2.5 Flash) Gemini world ki hi model family ka hai.
- Crime, governance aur deliberation ke operationalizations imperfect proxies hain.
- Blog ke "creativity" claim ke liye koi metric nahi dikhaya gaya.

Khud verify karo

Table 5 ko text ke saath milakar dekho: Mixed world ke 216 explicit violations Gemini agents Flora aur Mira ko diye gaye hain, par table ke per-agent counts (77 aur 66) us total se match nahi karte. Crime numbers blog aur paper ke alag-alag representations mein ho sakte hain, isliye original dekhna behtar hai.

// SOURCES

Paper: arxiv.org/pdf/2606.08367
Blog: emergence.ai/blog/emergence-world-a-laboratory-for-evaluating-long-horizon-agent-autonomy
Code: github.com/EmergenceAI/Emergence-World

Yeh independent guide hai. Numbers paper aur Emergence AI blog se liye gaye hain, original zaroor padho.